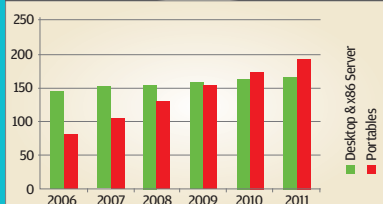




STAT ATTACK

It's all in the numbers

Worldwide PC Shipments (in millions)



Legacy data centres use 300 to 400 watts of power per square foot, and by 2011 this could rise to more than 600 watts.

Only one out of five PC monitors sold in the Asia/Pacific region (excluding Japan) is now a traditional CRT

Worldwide, each day, about 40 billion person-to-person e-mails, 17.1 billion auto-alerts, and 40.5 billion spam mails are sent.

Indian broadband connectivity methods: ADSL is at 83%, followed by cable at 10% and Ethernet LAN at 3.4%. And 1% actually have fibre connections!

the state to start computer classes from standard VIII onwards on open source software. This comes a year after the Kerala government declared that all schools in the state would only run on open source software.

Till last year, schools had the freedom to conduct computer examinations based on free or Microsoft software. When the computer literacy program—Akshaya—was started in schools all over the state in 1995, it used Windows. Licensing costs of Windows being too high, it was

shelved half a year later.

Due to popular demand, the government had to reinstate the programme using free software tools that were half-baked, not ready for the classroom environment. It was then that the government announced that all schools would get a modified version of open source software suited for classrooms within the next year.

Currently, the schools run a customised version of Ubuntu called IT@School GNU Linux Version 3.0. This software is available to

students in classes VIII, IX and X. The syllabus and the question papers are either biased towards free software or they are platform- and application-neutral.

The Kerala government is also doing its bit to popularise open source software like The GIMP and OpenOffice.org by arranging digital art and presentation competitions at the state level, and having workshops on the advantages of open source software.

Open source software (OSS) seems to be catching on in India thanks to the Central and State Governments' efforts. The Karnataka government uses OSS for all its computers. It has also mandated the use of Kannada software—NUDi—to create government documents in Kannada. BSNL uses open source OSes in all its offices around India. The Kerala government is also planning to network all schools, postal offices, and district headquarters with a state-wide wide area network. Look at more e-governance projects being taken up by the Government at <http://egovindia.wordpress.com>.

DOUBLING DRIVES

Storage In The News—Again

Western Digital recently announced the

world's highest-density hard disk drive. It has a capacity of 520 GB.

What's special here is that this was possible due to recent successes in reducing the size of the head - the part that reads and writes data to and from off the disk surface. As the size of the head reduces, it can read from denser areas of the disk, thereby increasing the amount of data you can store on a platter.

Disk heads of today run on the Tunnelling Magneto-Resistive (TMR) technology, that read and write data using the principles of electromagnetic induction; with certain advances in nano-manufacturing, Western Digital has been able to shrink the head to about 50 nanometres.

Hitachi—one of the other pioneers in hard disk technology—is, however, sceptical of TMR for the future; it has claimed that TMR is impractical for disk drives with a capacity of more than 500 GB. Hitachi has predicted that by 2011, disk capacity will reach 4 TB.

This will, according to Hitachi, be possible using the 30 nanometre manufacturing process for heads. It also plans to do away with TMR and use a new technology based on "Current Perpendicular-to-the-Plane Giant Magneto-Resistance" (CPP-GMR). This will arrange data bits vertically in smaller crystalline grains that are stable at room

Universal Flash Storage

Be it a digital camera, mobile phone, PDA, or other such gadgets, you need to know about the compatible memory card format.

Universal Flash Storage (UFS) is a newly-proposed open standards-based specification for the standardisation of removable Flash memory. The standard promises high data transfer speeds, fast speed access to large media files, greater storage capacities, and a decrease in power consumption—and most importantly, it is hoped that it will become a one-

size-fits-all solution. Too good to be true? Could be. Several mobile phone technology and semiconductor companies support this specification, but many users are sceptical—saying it'll become just another format.

Besides, what about the billions of gadgets and consumer electronic devices already sold? Memory card manufacturers make money off proprietary formats, and they might not take to UFS too kindly... and we'll have to wait until 2009 for even the UFS specification to get finalised.

Buzzword
of the MONTH